Let us consider an object of mass m and specific heat capacity s at temperature T. Let T_s be the temperature of the surroundings. If the temperature falls by a small amount dT in time dt, then the amount of heat lost is,

$$dQ = msdT \qquad (8.5)$$

Dividing both sides of equation (8.5) by dt

$$\frac{dQ}{dt} = \frac{msdT}{dt} \qquad (8.6)$$

From Newton's law of cooling

$$\frac{dQ}{dt} \propto -(T - T_s)$$

$$\frac{dQ}{dt} = - a (T - T_s) \qquad (8.7)$$

Where a is some positive constant.
From equation (8.6) and (8.7)

$$- a (T - T_s) = ms \frac{dT}{dt}$$

$$\frac{dT}{T - T_s} = -\frac{a}{ms} dt \qquad (8.8)$$

Integrating equation (8.8) on both sides,

$$\int \frac{dT}{T - T_s} = -\int \frac{a}{ms} dt$$

$$\ln (T - T_s) = -\frac{a}{ms} t + b_1$$

Where b_1 is the constant of integration. taking exponential both sides, we get

$$T = T_s + b_2 \, e^{-\frac{a}{ms}t} \qquad (8.9)$$

here $b_2 = e^{b_1}$ = constant

EXAMPLE 8.8

A hot water cools from 92°C to 84°C in 3 minutes when the room temperature is 27°C. How long will it take for it to cool from 65°C to 60°C?

The hot water cools 8°C in 3 minutes. The average temperature of 92°C and 84°C is 88°C. This average temperature is 61°C above room temperature. Using equation (8.8)

$$\frac{dT}{T - T_s} = -\frac{a}{ms} dt \text{ or } = \frac{dT}{dt} = -\frac{a}{ms}(T - T_s)$$

$$\frac{8^o C}{3\,\text{min}} = -\frac{a}{ms}(61^o C)$$

Similarly the average temperature of 65°C and 60°C is 62.5°C. The average temperature is 35.5°C above the room temperature. Then we can write

$$\frac{5^o C}{dt} = -\frac{a}{ms}(35.5^o C)$$

By diving both the equation, we get

$$\frac{\frac{8^o C}{3\,\text{min}}}{\frac{5^o C}{dt}} = \frac{-\frac{a}{ms}(61^o C)}{-\frac{a}{ms}(35.5^o C)}$$

$$\frac{8 \times dt}{3 \times 5} = \frac{61}{35.5}$$

$$dt = \frac{61 \times 15}{35.5 \times 8} = \frac{915}{284} = 3.22\,\text{min}$$

8.3 LAWS OF HEAT TRANSFER

8.3.1 Prevost theory of heat exchange

Every object emits heat radiations at all finite temperatures (except 0 K) as well as it absorbs radiations from the surroundings. For example, if you touch someone, they might feel your skin as either hot or cold.

A body at high temperature radiates more heat to the surroundings than it receives from it. Similarly, a body at a lower temperature receives more heat from the surroundings than it loses to it.

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